

Zhenyu Li

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Highlights

Highly Cited: ~2K citations in total of my publications as a **third-year** PhD student

Widely Impactful: ~2K stars on GitHub

Experienced: 2.5 years of industrial intern experience. 5+ years of research experience.

Prestigious affiliations: Meta, TikTok, DiDi, SenseTime Research

Keywords: computer vision; deep learning; depth estimation; diffusion model; 3D perception and understanding; multi-modality learning; semi-supervised learning; 3D foundation model; LLM; benchmarking

Experience

Research Intern

Meta

Zurich, Switzerland

Aug. 2025 - Feb. 2026

- Researched scalable egocentric body tracking algorithms for AR/VR (from both architecture design and data utilization aspects). It led to a **CVPR 2026 Highlight paper**.

Research Intern (*TopSeed* candidate)

TikTok

Beijing, China

Jan. 2025 - July 2025

- Researched benchmarking for depth estimation foundation models. The result was presented in the [paper](#).
- Collaborated the development for **DepthAnything 3 (ICLR 2026 Oral)**, the SoTA feed-forward 3D reconstruction method.

Elite Research Intern (Elite Camp 2022)

Didi Cargo (ADAS team)

Beijing, China

Sep. 2022 - Mar. 2023

- Researched semi-supervised algorithms for monocular 3D object detection. The solution improves the model performance with limited labeled data and was presented in the **ICRA 2025 paper**.

Research Intern

SenseTime Research (ADAS team)

Shanghai, China

Jan. 2022 - Jul. 2022

- Researched unsupervised domain adaptation algorithms for monocular 3D object detection. It led to an **ECCV 2024 paper**.
- Deployed the aforementioned unsupervised domain adaptation algorithm in an industrial project with GAC Group. Achieved excepted goals.
- Researched domain generalization algorithms for monocular 3D object detection (a follow-up work of the aforementioned paper). It was presented in the [paper](#).
- Collaborated with other team members and researched 3D object detection. It led to an **ICLR 2023 paper**.

Perception Algorithm Development Intern

SenseTime Research (ADAS team)

Shanghai, China

Mar. 2021 - Sep. 2021

- Built up a ReID dataset based on the ground-truth system, trained a ReID model, and developed the ReID model for the ADAS system.
- Built and deployed a multi-object tracking algorithm for the ADAS system (C++), including importing appearance representation from the ReID model and adopting the cascade association strategy.
- Researched multi-modal contrastive learning algorithms for spatial-aware visual representations to benefit 3D-related downstream tasks. It led to an **AAAI 2022 paper**.
- Collaborated with other team members and researched 3D object detection. It led to an **IJCAI 2022 paper** and an **ECCV 2022 paper**.

Selected First Author Publications

- **EgoPoseFormer V2: Accurate Egocentric Human Motion Estimation for AR/VR.** *CVPR 2026 (Highlight)*. Zhenyu Li, Sai Kumar Dwivedi, Filip Maric, Carlos Chacón, Nadine Bertsch, Filippo Arcadu, Tomas Hodan, Michael Ramamonjisoa, Peter Wonka, Amy Zhao, Robin Kips, Cem Keskin, Anastasia Tkach, Chenhongyi Yang. [Project link](#).
- **PatchRefiner V2: Fast and Lightweight Real-Domain High-Resolution Metric Depth Estimation.** *ICLR 2026*. Zhenyu Li, Wenqing Cui, Shariq Farooq Bhat, Peter Wonka. [Project link](#).
- **Amodal Depth Anything: Amodal Depth Estimation in the Wild.** *ICCV 2025*. Zhenyu Li, Mykola Lavreniuk, Jian Shi, Shariq Farooq Bhat, Peter Wonka. [Project link](#).
- **PatchRefiner: Leveraging Synthetic Data for Real-Domain High-Resolution Monocular Metric Depth Estimation.** *ECCV 2024*. Zhenyu Li, Shariq Farooq Bhat, Peter Wonka. [Project link](#).
- **PatchFusion: An End-to-End Tile-Based Framework for High-Resolution Monocular Metric Depth Estimation.** *CVPR 2024*. Zhenyu Li, Shariq Farooq Bhat, Peter Wonka. [Project link \(1k+ stars on GitHub\)](#).
- **Unsupervised Domain Adaptation for Monocular 3D Object Detection via Self-Training.** *ECCV 2022*. Zhenyu Li, Zehui Chen, Ang Li, Liangji Fang, Qinhong Jiang, Xianming Liu, Junjun Jiang.
- **SimIPU: Simple 2D Image and 3D Point Cloud Unsupervised Pre-Training for Spatial-Aware Visual Representations.** *AAAI 2022*. Zhenyu Li, Zehui Chen, Ang Li, Liangji Fang, Qinhong Jiang, Xianming Liu, Junjun Jiang, Bolei Zhou, Hang Zhao. [Project link](#).
- **BinsFormer: Revisiting Adaptive Bins for Monocular Depth Estimation.** *TIP*. Zhenyu Li, Xuyang Wang, Xianming Liu, Junjun Jiang. [Project link](#).
- **DepthFormer: Exploiting Long-Range Correlation and Local Information for Accurate Monocular Depth Estimation.** *MIR*. Zhenyu Li, Zehui Chen, Xianming Liu, Junjun Jiang. [Project link](#).
- **CriterionAlign: Criterion-Centric Rationale Alignment for Code Preference Judging.** *Under Review*. Zhenyu Li, Aleksandar Cvejic, Zehui Chen, Peter Wonka. [Project link](#).

Selected Co-Author Publications

- **Depth Anything 3: Recovering the Visual Space from Any Views.** *ICLR 2026 Oral*. Haotong Lin, Sili Chen, Junhao Liew, Donny Y. Chen, Zhenyu Li, Guang Shi, Jiashi Feng, Bingyi Kang. [Project link](#).
- **Any Resolution Any Geometry: From Multi-View To Multi-Patch.** *CVPR 2026*. Wenqing Cui, Zhenyu Li, Mykola Lavreniuk, Jian Shi, Ramzi Idoughi, Xiangjun Tang, Peter Wonka. Bingyi Kang. [Project link](#).
- **DissolveStereo: Coarse Depth Injection for Zero-Shot Stereo Video Generation.** *SIGGRAPH, TOG 2026*. Jian Shi, Qian Wang, Zhenyu Li, Wenqing Cui, Ramzi Idoughi, Peter Wonka. [Code](#).
- **Lari: Layered Ray Intersections for Single-View 3D Geometric Reasoning.** *ICML 2026*. Rui Li, Biao Zhang, Zhenyu Li, Federico Tombari, Peter Wonka. [Project link](#).
- **Learning from Noisy Data for Semi-Supervised 3D Object Detection.** *ICCV 2023*. Zehui Chen, Zhenyu Li, Shuo Wang, Dengpan Fu, Feng Zhao.
- **BEVDistill: Cross-Modal BEV Distillation for Multi-View 3D Object Detection.** *ICLR 2023*. Zehui Chen, Zhenyu Li, Shiquan Zhang, Liangji Fang, Qinhong Jiang, Feng Zhao. [Code](#).
- **Deformable Feature Aggregation for Dynamic Multi-Modal 3D Object Detection.** *ECCV 2022*. Zehui Chen, Zhenyu Li, Shiquan Zhang, Liangji Fang, Qinhong Jiang, Feng Zhao. [Code](#).

Projects

Monocular Depth Estimation Toolbox

[Link](#) 

- Developed several monocular depth estimation methods with reproduced results for fair comparisons
- **900+** stars on [GitHub](#)

Education

King Abdullah University of Science and Technology (KAUST) <i>PhD in Computer Science, Advisor: Prof. Peter Wonka</i>	Thuwal, Saudi Arabia <i>Sep. 2023 - Present</i>
Harbin Institute of Technology (HIT) <i>MS in Computer Science, Advisor: Prof. Junjun Jiang. Program rank - 2/80+</i>	Harbin, China <i>Sep. 2021 - Jul. 2023</i>
Harbin Institute of Technology (HIT) <i>BS in Computer Science, Advisor: Prof. Junjun Jiang. GPA - 89.58/100</i>	Harbin, China <i>Sep. 2017 - Jul. 2021</i>

Awards

Dean's List Award , award to top students by KAUST CEMSE for their academic achievements	2024
ICCV VCL 2023 Multitask Learning for Robustness Challenge , 1st place	2023
ECCV SSLAD 2022 3D Object Detection Challenge , 3rd Place	2022
ECCV Mobile AI & AIM 2022 Monocular Depth Estimation Challenge , 2nd Place	2022
China National Scholarship	2022

Skills

Technical Skills:

Languages (Years of Experience, Last Used): Python (6 years, current), C (7 years, 2018), C++ (3 years, 2021), Java (6 years, 2019).

Tools & Frameworks: Pytorch (profecient), Tensorflow (familiar), OpenMMLab Toolbox, *etc.*

Soft Skills:

Languages: Chinese (Mother Tongue), English (Fluent)

Teamwork: Can collaborate with team members with diverse backgrounds, resulting in enhanced efficiency.

Problem-solving: Proficient in identifying issues and implementing solutions that exceed project objectives.

Adaptability for Projects: Can take advantages of previous experiences to adapt to various new projects.

Time Management: Expert in prioritizing tasks and managing schedules to maximize productivity.